Adapting Pretrained Models: Fine-Tuning, Visual Grounding, and Test-Time Compute

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Abstract

We study how parameter constraints, multimodal alignment, and test-time computation affect small pretrained language and vision models across three empirical tasks. First, we fine-tune Qwen3-0.6B on mathematics using Full FT, L2-SP, LoRA, and OPLoRA: full-weight runs achieve the largest GSM8K gains (up to +21.0 percentage points [pp]), whereas rank-16 adapters trade 3.0–3.3 pp of peak math gain for 70–73% lower peak GPU memory and stronger out-of-domain retention. Second, we train a SigLIP–Qwen3-0.6B vision-language model through a three-stage curriculum; visual sensitivity rises sharply during Stage 2 instruction tuning, and the final model outperforms complementary-image controls by +55.5 pp on grounding and +29.5 pp on VQA, while its strict-metric drop without an image partly reflects full-sentence formatting and counting and pair consistency remain limited. Third, on Qwen2.5 math and multimodal rollouts alongside a 50-question live study on Qwen2.5-3B, large-budget errors stem primarily from misselecting existing correct candidates: rank-weighted consensus significantly improves MATH accuracy by +1.8 pp over self-consistency, adaptive stopping cuts token use by 52–64% with little or no accuracy loss, and live parallel or refinement decoding shows no significant gain while parallel sampling increases truncation. Code: `https://github.com/FaaizKhan1999/atml-assignment3`

1 Introduction

Pretrained language and vision models are strong starting points for downstream tasks, but adapting and using them effectively raises three practical questions. First, when fine-tuning a small language model on a specialized domain like mathematics, parameter updates can overwrite existing capabilities [1]; we must understand how weight penalties, low-rank adapters, and subspace protection trade target-domain adaptation against retention and compute cost [2, 3]. Second, when connecting a frozen vision encoder [4] to a language model [5], predictions can rely on text shortcuts and language priors rather than visual evidence [6]. Third, when weights are fixed at inference time, repeated sampling increases candidate coverage, but it remains unclear how best to select among candidates or when extra rollouts justify their token cost [7–11].

2 Experimental Setup and Methods

2.1 Mathematical Fine-Tuning and Parameter-Restricted Adaptation

Model, Data, and Training. We initialize five runs from `Qwen/Qwen3-0.6B` [12] in PyTorch and Transformers [13, 14] (seed 6304, `enable_thinking=False`, leading response newlines stripped) on the first 40,000 eligible MetaMathQA [15] examples after length and GSM8K-overlap filtering on `query` and `original_question` (Appendix Table 8). Each run trains for one epoch (1,250 updates,

effective batch 32) using AdamW, cosine decay, gradient checkpointing, FP16 mixed precision, and response-only cross-entropy. All five runs executed on one Kaggle T4 $\times$ 2 node across two concurrent per-GPU queues, affecting wall-clock times.

Configurations and Evaluation. We evaluate seven conditions (shared baseline, Full FT midpoint at update 625, and five final models at update 1,250): Full FT (LR $= 2 \times 10^{-5}$); L2-SP [2] ($\lambda = 10^{-3}$, reference θ_0 in pinned CPU memory); LoRA-4 and LoRA-16 [16] (LR $= 2 \times 10^{-4}$, $(r, \alpha) \in \{(4, 8), (16, 32)\}$ on all seven linear projections per layer); and OPLoRA-16 [3] ($r = 16, \alpha = 32, k = 128$, projecting adapters outside the top-k FP32 singular subspaces of W_0; merge and projection checks in Appendix Table 12). Using `lm-evaluation-harness` [17] in zero-shot FP16, we evaluate GSM8K test [18] (greedy, 1,024-token cap) and likelihood accuracy on HellaSwag [19], ARC-Easy [20] (both length-normalized), and BoolQ [21] (Appendix Table 9). On `down_proj` in layers 0, 13, and 27, we compute the protected-subspace energy ratio ρ_k, relative update norm η, and FP32 activation shift $R(z)$ on cached baseline inputs across 32 GSM8K and 32 HellaSwag probes. We also greedily generate 512-token responses on 24 non-math prompts [22] and manually annotate a fixed 42-response subset (unblinded).

2.2 Vision-Language Interface and Grounding Evaluation

Architecture, Data, and Curriculum. Following Liu et al. [5], we cache 196 normalized FP16 patch features from frozen `siglip-base-patch16-224` [4] and map them into `Qwen/Qwen3-0.6B` [12] via a two-layer MLP connector ($768 \to 1024 \to 1024$) scaled by a fixed training-derived calibration factor (0.0555) to match text-embedding RMS. From image-disjoint COCO 2014 [23, 24] and VQAv2 [6] splits (Appendix Table 16), we build COCO captions, programmatic presence/counting pairs filtered for label quality, and natural VQAv2 complementary pairs ($\geq 8/10$ consensus). We train sequentially over 4,500 updates (effective batch 16, example-weighted cross-entropy; costs in Appendix Table 20): Stage 1 (2,250 updates) trains only the connector on captions; Stage 2 (1,500 updates) adds attention LoRA ($r = 16, \alpha = 32$) [16] on programmatic grounding with caption replay; and Stage 3 (750 updates) specializes on VQAv2 with replay of all prior tasks.

Evaluation and Controls. We log dev caption loss, answer NLL, and the complementary swap penalty ΔNLL across training. Final Stage 3 evaluation (greedy, 16-token cap) tests Correct, Complementary swap, Mismatched (unrelated image), and No image conditions using normalized exact match (grounding), VQA soft consensus accuracy [25], pair accuracy, answer NLL, and 4,000-resample pair-bootstrap 95% CIs. Because no-image responses default to full sentences, we add a supplementary non-protocol lead-answer accuracy scoring the first yes/no word or number (excluding "other" questions).

2.3 Test-Time Compute: Sampling, Selection and Adaptive Inference

Rollouts, Grading, and Signals. We analyze 32 supplied rollouts per question from `Qwen2.5-3B-Instruct` [26] on 1,000 MATH problems [27] and `Qwen2.5-VL-3B-Instruct` [28] on 1,000 MathVista `testmini` problems [29], split (seed 6304) into 200 dev and 800 test questions across 10 rollout orders for $N \in \{1, 2, 4, 8, 16, 32\}$. Answers are grouped without gold references: on MATH, `math-verify` [30] groups non-empty strings (80 timed-out comparisons counted as non-equivalent; Appendix Table 21) and maps empty extractions to `NO_ANSWER`; on MathVista, metadata normalization maps missing or unparseable extractions to `NO_ANSWER` (lenient scorer check in Appendix Table 22). `NO_ANSWER` participates in voting and cost accounting but is graded incorrect. Selection, calibration, and stopping never access gold answers or unrevealed rollouts. Per rollout, we compute mean log-probability L_i and self-certainty K_i [9], plus MathVista visual engagement V_i and VisConf $Z_i = V_i C_i$ [10]. We compare `FirstAns`, self-consistency [7], highest-score selection, shortest-length control, and rank-weighted consensus (RWC), with 1,000-resample paired question bootstrap 95% CIs at $N = 32$.

Adaptive Stopping and Live Generation. We evaluate Agreement-based (unique non-`NO_ANSWER` plurality $\geq \tau$) and Quality-aware (additionally requiring self-consistency and RWC winners to match with RWC margin $\geq \gamma$) stopping [8] at $n \in \{4, 8, 16\}$ before a forced stop at 32, freezing dev-selected thresholds for test. On 50 held-out MATH questions using live FP16 `Qwen2.5-3B-Instruct` on a Kaggle T4 (same sampling as the supplied rollouts, batch size 1, 3,072-token total budget), we compare a Single response (3,072 cap), Parallel $\times 3$ (three 1,024-token rollouts with

Table 1: Zero-shot benchmark evaluation across mathematical and general reasoning tasks. While full-parameter fine-tuning and L2-SP achieve the largest in-domain mathematical gains at the expense of specific reading comprehension skills, low-rank adapters—particularly higher-rank LoRA and OPLoRA—deliver substantial mathematical improvements alongside positive transfer to out-of-domain benchmarks.

Condition	GSM8K (%)	HellaSwag (%)	BoolQ (%)	ARC-E (%)	Mean non-math Δpp	GSM8K unextr. %	GSM8K trunc. %
Baseline	43.4	47.3	64.4	56.0	–	0.0	0.8
Full FT (mid)	56.9 (+13.6)	45.7 (-1.6)	62.6 (-1.8)	62.1 (+6.1)	+0.89	0.0	1.1
Full FT	64.4 (+21.0)	45.9 (-1.4)	61.5 (-2.8)	63.7 (+7.7)	+1.14	0.0	0.2
Full FT + L2-SP	64.2 (+20.8)	46.6 (-0.7)	61.6 (-2.8)	63.7 (+7.7)	+1.38	0.0	0.5
LoRA-4	59.9 (+16.5)	47.5 (+0.1)	65.4 (+1.0)	63.1 (+7.1)	+2.75	0.0	1.0
LoRA-16	61.0 (+17.7)	47.5 (+0.1)	71.7 (+7.4)	64.4 (+8.3)	+5.28	0.0	0.9
OPLoRA-16	61.3 (+18.0)	47.3 (+0.0)	71.8 (+7.4)	63.9 (+7.9)	+5.09	0.1	0.7

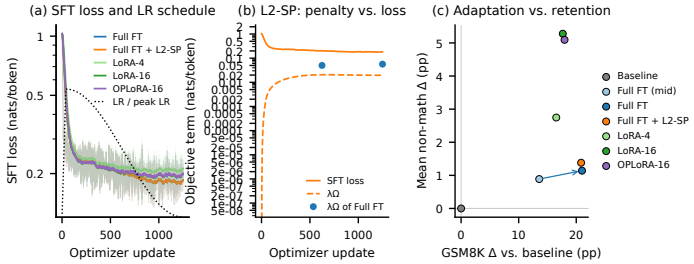


Figure 1: Training dynamics, regularization, and the adaptation–retention trade-off. L2-SP restricts parameter drift without sacrificing Full FT's mathematical gains, whereas adapters prioritize out-of-domain generalization over peak target-task accuracy.

self-consistency), and Sequential refinement [11] (shared first 1,024-token rollout plus a 2,048-token critique-and-revision pass without oracle feedback).

3 Results and Analysis

3.1 Specialization, Retention and Restrictions on Adaptation

Specialization, Progression, and Retention. Fine-tuning raises GSM8K accuracy by +16.5 to +21.0 percentage points (pp) over the 43.4% baseline without causing catastrophic forgetting on aggregate non-math benchmarks (Table 1; Appendix Table 9) [17]. Extraction artifacts are ruled out, though single-run scores carry roughly ± 2.6 pp of 95% binomial sampling noise on GSM8K. Full FT acquires two-thirds of its math gain by the midpoint while keeping mean non-math change positive. However, aggregate retention masks opposite mechanisms on BoolQ (Appendix Table 10): Full FT's -2.8 pp raw accuracy drop reflects a shift toward predicting "no" ("yes" rate falling from 57.7% to 31.5%) even as its class-balanced accuracy rises. Conversely, LoRA-16 and OPLoRA-16 gain +7.4 pp on BoolQ without a label-bias shift—improving on both classes—driving their higher mean non-math gains [1].

Controlled Comparisons of Update Restrictions. First, adding L2-SP [2] to Full FT cuts parameter displacement $\|\theta - \theta_0\|_2$ from 10.65 to 6.26 (Figure 1, Table 2), while matching Full FT on GSM8K and non-math tasks (within single-run noise). Second, quadrupling rank from LoRA-4 to LoRA-16 [16] improves GSM8K (+16.5 to +17.7 pp) and raises mean non-math change from +2.75 to +5.28 pp, though rank-16 adapters still trail Full FT by 3.0–3.3 pp on GSM8K, consistent with low-rank updates learning less in-domain while perturbing baseline behavior less [1]. Third, OPLoRA-16 matches LoRA-16 [3] at identical rank and initialization, indicating that orthogonal subspace protection does not noticeably alter benchmark scores at this scale.

Update Geometry, Activation Shifts, and Resource Trade-Offs. On `down_proj` (Figure 2; Appendix Tables 12 and 11), unconstrained updates place $\rho_k \approx 0.13$ of their energy inside the top-k starting subspace—matching the random-direction baseline $k/d_{\text{out}} = 0.125$—whereas OPLoRA-16 drives ρ_k to 0.0000 [3] at a similar relative update norm. Local activation shifts $R(z)$ are layer-dependent: in layers 0 and 13, updates respond more strongly to GSM8K than HellaSwag probes, whereas in layer 27 the ratio reverses across all methods, though cached-input $R(z)$ probes measure isolated layer responses rather than causal end-to-end effects. Computationally (Table 2), adapters

Table 2: Computational resource footprints, parameter efficiency, and weight-space displacement across fine-tuning regimes. Low-rank adaptation drastically reduces peak GPU memory and trainable parameter counts compared to full fine-tuning, though orthogonal projection introduces additional training latency and subspace precomputation overheads.

Method	Trainable params	Train time (h)	Peak GPU mem. (GB)	Recovery ckpt	Exported weights	$\|\theta - \theta_0\|_2$	Final $\lambda\Omega$	Method-specific overhead
Full FT	596.05M (100.00%)	2.36	11.95	6.66 GB	2.22 GB	10.65	(0.0567)	–
Full FT + L2-SP	596.05M (100.00%)	2.46	11.95	6.66 GB	2.22 GB	6.26	0.0196	θ_0 copy 2.22 GB (pinned CPU); penalty 5.6 min
LoRA-4	2.52M (0.42%)	2.68	3.14	29.3 MB	9.7 MB	–	–	export 0.0 s
LoRA-16	10.09M (1.67%)	2.63	3.25	115.9 MB	38.5 MB	–	–	export 0.1 s
OPLoRA-16	10.09M (1.67%)	3.50	3.56	115.9 MB	346.6 MB	–	–	SVD+QR 38.8 s; bases 308.0 MB; export 0.6 s

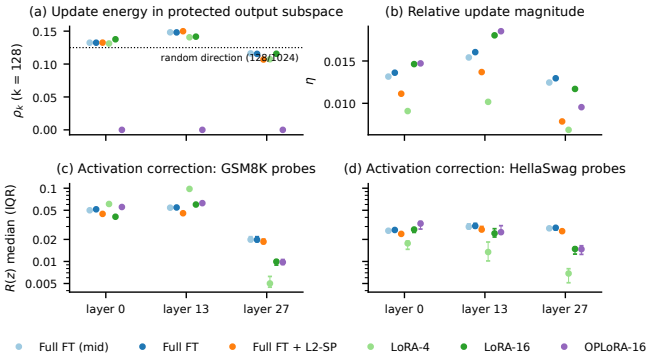


Figure 2: Subspace update energy, relative update norm, and activation shift across layers. Unconstrained updates match the random-subspace baseline, whereas OPLoRA eliminates protected-subspace interference while inducing comparable activation shifts.

Table 3: Qualitative human evaluation of generation coherence, formatting compliance, and failure modes across a small response sample. Unconstrained full fine-tuning reduces mean coherence by introducing severe repetition and unprompted mathematical framing, whereas L2-SP avoids severe repetition and low-rank adapters improve coherence while largely avoiding mathematical framing on general prompts, with zero inappropriate content observed across all conditions.

Condition	Format: yes/partial/no	Mean coherence (1–3)	Repetition: some/severe	Math framing	Truncated	Inappropriate
Baseline	2/2/2	2.33	0/0	0/6	0/6	0/6
Full FT (mid)	2/2/2	1.83	2/2	4/6	2/6	0/6
Full FT	3/2/1	2.17	3/2	3/6	2/6	0/6
Full FT + L2-SP	3/2/1	2.17	3/0	4/6	0/6	0/6
LoRA-4	3/1/2	2.50	0/1	0/6	0/6	0/6
LoRA-16	3/2/1	2.50	2/1	0/6	1/6	0/6
OPLoRA-16	3/1/2	2.67	1/0	1/6	0/6	0/6

train just 0.42–1.67% of parameters and cut peak GPU memory by 70–74% [16], while L2-SP adds pinned-CPU reference overhead [2] and OPLoRA-16's SVD/QR precomputation and per-step projections increase runtime (under concurrent two-queue execution).

Qualitative Behavior Beyond Benchmark Scores. Unblinded annotation of the 42-response subset ($n = 6$ per condition; Table 3) and automated flags on all 168 outputs (Appendix Table 13; aligned excerpts in Appendix Tables 14 and 15) expose generative side effects that teacher-forced likelihood benchmarks miss [22]. Full FT lowers coherence through severe repetition and unprompted mathematical framing on non-math prompts; L2-SP eliminates severe repetition but retains math framing, whereas adapters largely avoid math framing and maintain slightly higher coherence (with zero inappropriate content anywhere, though small differences over six unblinded responses reflect single-sample shifts).

3.2 Learning a Visual Interface and Measuring Visual Grounding

Curriculum Progression and Stage Transitions. In Stage 1 (caption alignment; Figure 3; Appendix Tables 16 and 17) [5], training only the connector lowers dev caption loss and reduces correct-image answer NLL overall, though answer NLL rebounds after update 1,000 (consistent with prolonged caption training favoring descriptive phrases over short answers) and the swap penalty ΔNLL stays

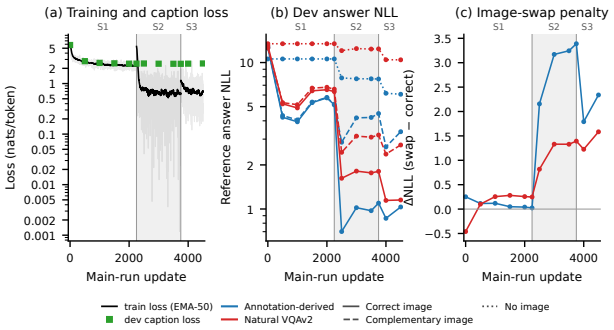


Figure 3: Multi-stage training trajectories and complementary-image swap penalties. Caption alignment reduces overall answer loss before rebounding, whereas instruction tuning sharply lowers answer loss and widens the correct-over-complementary likelihood gap.

Table 4: Final stage-three evaluation under correct, complementary, mismatched, and no-image conditions. The model achieves its highest accuracy and lowest answer loss with the correct image compared to complementary or mismatched inputs, while removing the image causes the model to generate identical full-sentence responses per prompt, which reduces strict exact-match accuracy far more than lead-answer accuracy.

Suite	Image	Accuracy % [95% CI]	By type %	Pair acc. %	Identical %	Empty %	Ref. NLL	ms/ans.	Lead-ans. %†
Annotation-derived (64 pairs)	Correct	66.4 [57.0, 75.0]	89 / 44	50.0	20.3	0.0	0.93	42	66.4
	Complementary	10.9 [6.2, 16.4]	11 / 11	0.0	20.3	0.0	4.02	30	10.9
	Mismatched	37.5 [30.5, 45.3]	58 / 17	9.4	64.1	0.0	2.38	30	37.5
	No image	0.0 [0.0, 0.0]	0 / 0	0.0	100.0	0.0	6.55	111	25.0
Natural VQAv2 (96 pairs)	Correct	56.6 [50.0, 63.1]	61 / 48 / 58	27.1	40.6	0.0	1.40	35	55.9
	Complementary	27.1 [20.9, 33.5]	48 / 19 / 11	5.2	40.6	0.0	2.85	35	36.5
	Mismatched	35.4 [29.1, 42.1]	59 / 30 / 16	8.3	56.2	0.0	2.34	39	47.0
	No image	1.0 [0.0, 2.6]	0 / 4 / 0	0.0	100.0	0.0	9.52	123	29.2

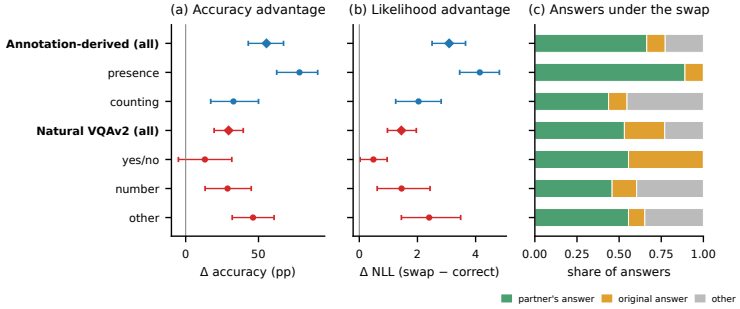


Figure 4: Bootstrap analysis of visual sensitivity and prediction shifts under complementary swaps. Correct images yield significant accuracy and likelihood gains, and swapping images shifts over half of VQA answers to the partner's reference.

near zero. Enabling attention LoRA and programmatic grounding pairs in Stage 2 sharply lowers answer NLL and drives ΔNLL up to $+3.39$ on grounding and $+1.39$ on VQA while caption loss stays flat. In Stage 3, VQAv2 specialization raises VQA ΔNLL to $+1.58$, while grounding ΔNLL moderates to $+2.34$.

Visual Grounding Across Question Suites and Controls. Relative to the correct-image baseline, the Stage 3 model shows strong visual sensitivity, with larger advantages on programmatic grounding than on natural VQAv2 (Table 4; Appendix Tables 18 and 19) [6, 25]. Swapping to the complementary partner image—keeping question wording and visual-token count fixed while changing the reference answer—cuts accuracy by 55.5 pp on grounding and 29.5 pp on VQA. Unrelated mismatched images yield intermediate accuracy and answer NLL because random scenes occasionally share the original answer by chance, whereas complementary partners are constructed to differ. Removing visual tokens collapses strict accuracy to 0.0–1.0%; however, our supplementary non-protocol lead-answer metric reaches 25.0% on grounding and 29.2% on scored yes/no and number VQA prompts, showing that the strict no-image drop combines missing visual evidence with a shift toward full-sentence outputs.

Consistency Across Likelihood, Generation, and Pair Diagnostics. Correct images significantly reduce answer NLL over complementary swaps ($+3.09$ on grounding; $+1.45$ on VQA; Figure 4,

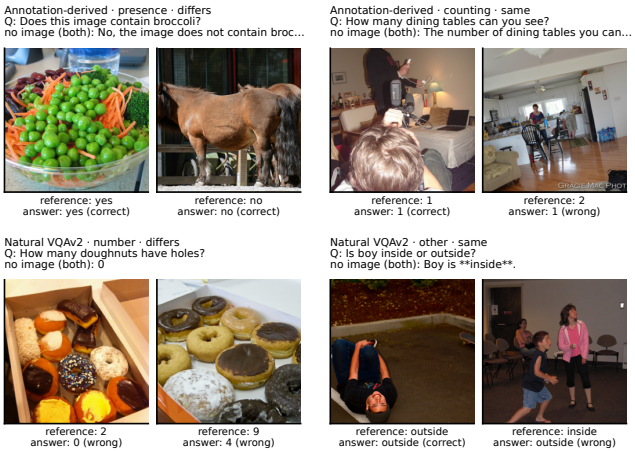


Figure 5: Predictions across four representative complementary pairs and no-image controls. The model resolves binary presence when given images and reverts to full sentences without an image on most prompts, but misses references on counting and spatial pairs.

Table 5: Comparison of candidate aggregation and selection rules across mathematical and multimodal benchmarks. Single-candidate selection heuristics and shortest-response controls perform worse than standard majority voting, whereas confidence-weighted voting yields a statistically significant gain on MATH and modest point-estimate improvements on MathVista whose confidence intervals include or touch zero.

Method	MATH N=8	MATH N=32	MATH Δ vs SC @32 [95% CI]	MathVista N=8	MathVista N=32	MathVista Δ vs SC @32 [95% CI]
FirstAns	64.4	64.4	-9.4 [-11.1, -7.8]	53.6	53.6	-8.1 [-10.0, -6.2]
Self-consistency	71.8	73.8	–	59.9	61.7	–
Max mean log-prob	67.5	64.0	-9.8 [-12.4, -7.2]	51.9	48.1	-13.6 [-17.0, -10.3]
Max self-certainty	70.1	70.6	-3.2 [-5.6, -0.8]	54.8	54.6	-7.1 [-10.2, -3.8]
Self-certainty + RWC	72.9	75.5	+1.8 [+0.7, +2.8]	60.0	62.5	+0.8 [-0.6, +2.1]
Shortest (length control)	64.6	62.4	-11.4 [-14.1, -9.0]	54.6	53.8	-7.8 [-11.0, -4.9]
Max vision	–	–	–	54.3	53.6	-8.1 [-11.3, -4.9]
Vision + RWC	–	–	–	60.1	62.9	+1.2 [-0.3, +2.8]
Max VisConf	–	–	–	56.3	57.8	-3.9 [-7.0, -0.9]
VisConf + RWC	–	–	–	60.8	63.1	+1.4 [+0.0, +3.0]
Max VisConf (tokenwise)	–	–	–	55.7	56.2	-5.4 [-8.4, -2.4]
VisConf tokenwise + RWC	–	–	–	60.9	62.6	+1.0 [-0.4, +2.4]
Oracle coverage (upper bound)	84.3	91.4	–	82.7	89.5	–

Table 4) [6]. Under complementary swaps, 53.1% of VQA answers switch to the partner's reference while 24.0% retain the original answer (consistent with language priors, ungrounded persistence, or failing to resolve visual differences), whereas on grounding the partner-match rate equals correct-image accuracy by construction. Yet strict pair accuracy—answering both images in a pair correctly—is much lower than single-image accuracy, and removing the image yields 100% identical predictions across paired prompts, confirming that within-pair divergence under visual conditions is strictly image-driven.

Capabilities, Qualitative Failure Modes, and Computational Cost. Representative pairs—the first pair in ID order per question type where answers differ and the first where answers match (Figure 5; all 10 pairs across the five question types in Appendix Figure 8)—show strong binary presence detection (`yes/no` on broccoli) alongside errors on counting and spatial queries. Computationally (Appendix Table 20, Table 4), caching SigLIP features [4] keeps peak training memory at 1.54–1.56 GB across all stages [5], with total training taking 1.55 h and greedy inference taking 30–42 ms per answer with an image (111–123 ms without).

3.3 Allocating Test-Time Compute: Sampling, Selection and Stopping

Exploration Opportunity vs. Selection Bottlenecks. From $N = 1$ (`FirstAns`) to $N = 32$, oracle coverage—an upper bound using gold answers rather than a deployable selector—rises substantially (Figure 6) [7, 9]. Yet deployable selectors recover less than half of this gain: at $N = 32$, standard self-consistency leaves selection gaps of 17.6 pp on MATH and 27.8 pp on MathVista below oracle coverage while gains largely saturate by $N = 8$ (Table 5).

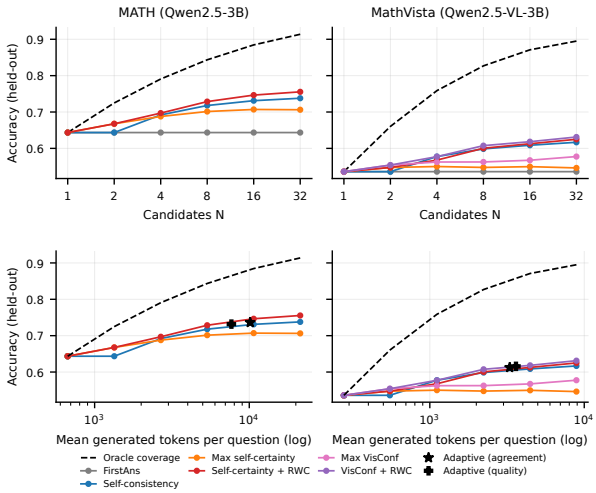


Figure 6: Accuracy scaling across candidate budgets and token costs for fixed and adaptive policies. Self-consistency leaves a wide gap to oracle coverage, while adaptive stopping cuts token use with minimal accuracy loss.

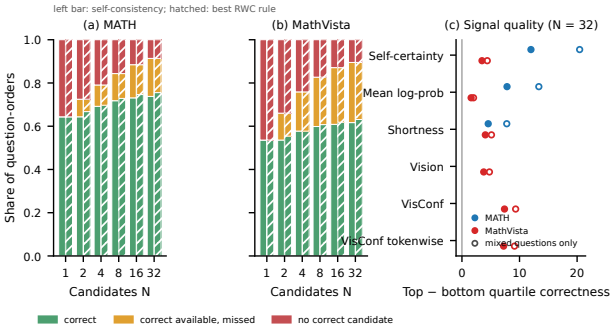


Figure 7: Candidate generation versus selection failures and quartile-based signal discrimination. Errors shift from generation misses at small budgets to selection failures at larger budgets, where domain-matched confidence signals discriminate best.

Separating Quality Signals from Consensus Aggregation. Controlled comparisons at $N = 32$ (Table 5; Appendix Tables 25–24) show that consensus aggregation matters far more than single-rollout score maximization [7]. Holding the quality signal fixed while switching from best-of-N (max score) to rank-weighted consensus (RWC) jumps accuracy from 70.6% to 75.5% for self-certainty on MATH and from 57.8% to 63.1% for VisConf on MathVista; every single-best selector and the shortest-response control trail uniform voting. Replacing uniform voting with RWC yields a significant $+1.8$ pp gain on MATH using self-certainty ($[+0.7, +2.8]$) [9], while holding RWC fixed on MathVista and varying the signal puts VisConf highest, though the MathVista intervals touch or include zero [10].

Failure Decomposition and Signal Discrimination. As N grows, errors shift from lacking any correct candidate at small budgets to misselecting an available correct candidate from $N = 16$ on MATH and $N = 8$ on MathVista (Figure 7a,b). Within-question quartile gaps on mixed questions at $N = 32$ show why domain-matched signals guide RWC best (Figure 7c; Appendix Table 26 and Figure 10): top-minus-bottom quartile correctness is largest for self-certainty on MATH ($+20.4$ pp) [9] and VisConf on MathVista ($+9.3$ pp, roughly doubling self-certainty or vision alone) [10]. Within MathVista rollouts, token confidence and visual attention correlate negatively (median $r = -0.21$); this negative covariance lowers tokenwise fusion scores Z_i^{token} relative to trajectory-level Z_i, while in practice yielding slightly lower accuracy and nearly identical quartile discrimination.

Live Breadth vs. Sequential Refinement. On the 50-question live MATH study under a 3,072-token cap (Table 6), accuracy differences are statistically inconclusive given wide intervals: charging the shared 1,024-token initial rollout to both multi-call arms, parallel $\times 3$ and sequential refinement [11] consume 2.3–2.5$\times$ as many tokens. Splitting the allowance into three 1,024-token parallel calls raises truncation from 4% to 16% [7], whereas refinement keeps truncation at 5% but yields zero net gain,

Table 6: End-to-end latency, token overhead, and accuracy for live single-pass, parallel, and iterative refinement decoding. Under strict generation budgets, naive parallel sampling and multi-turn self-revision increase computational latency and truncation rates without improving accuracy, as beneficial revisions are offset by self-induced corruptions.

Strategy	Accuracy %	Δ vs single [95% CI]	Input tok.	Output tok.	Total tok.	s / question	Truncated calls %	No extraction %
Single response ($\leq$3,072)	66.0	–	108	734	843	36.5	4.0	4.0
Parallel $\times$3 ($\leq$1,024) + SC	62.0	-4 [-16, +6]	325	1821	2146	89.2	16.0	16.0
Initial + critique-revision ($\leq$2,048)	64.0	-2 [-12, +10]	837	1117	1955	55.4	5.0	5.0

Table 7: Performance and stopping distributions of agreement- and quality-based adaptive early-stopping policies. Both adaptive policies halt at the minimum rollout threshold on a majority of question-order instances to cut average rollout counts by more than half, showing no statistically significant accuracy loss relative to full-budget voting except for a small drop under the quality-aware policy on MATH.

Dataset	Policy	Setting (dev)	Accuracy %	Δ vs SC@32 [95% CI]	Tokens / q.	Rollouts	Stops at 4 / 8 / 16 / 32 (%)
MATH	Fixed SC, N = 4	–	69.2	–	2680	4	–
	Fixed SC, N = 8	–	71.8	–	5344	8	–
	Fixed SC, N = 16	–	73.1	–	10678	16	–
	Fixed SC, N = 32	–	73.8	–	21347	32	–
	Adaptive: agreement	τ = 0.625	73.6	-0.1 [-0.5, +0.2]	10140	11.8	66 / 6 / 1 / 27
	Adaptive: quality	τ = 0.5, γ = 0.0	73.2	-0.6 [-1.2, -0.1]	7663	9.0	76 / 6 / 1 / 16
MathVista	Fixed SC, N = 4	–	57.8	–	1112	4	–
	Fixed SC, N = 8	–	59.9	–	2230	8	–
	Fixed SC, N = 16	–	60.9	–	4462	16	–
	Fixed SC, N = 32	–	61.7	–	8898	32	–
	Adaptive: agreement	τ = 0.5	61.3	-0.4 [-1.6, +0.7]	3290	9.9	71 / 8 / 2 / 19
	Adaptive: quality	τ = 0.5, γ = 0.0	61.6	-0.1 [-1.1, +0.9]	3613	10.9	65 / 10 / 3 / 22

trading 1 wrong→correct recovery against 1 correct→wrong corruption across 6 changed answers [31, 32].

Adaptive Early Stopping and Failure Modes. With dev-selected thresholds frozen for evaluation (Appendix Table 27), agreement and quality-aware stopping cut mean token cost by 52.5–64.1% (Table 7) [8, 10]. Three policies match fixed-32 self-consistency with CIs spanning zero while achieving higher point-estimate accuracy than fixed $N = 8$ or $N = 16$ budgets at comparable cost (Figure 6); only MATH quality-aware stopping incurs a small significant loss (-0.6 pp $[-1.2, -0.1]$). Retrospectively (Appendix Figure 11), quality-aware stopping halts at 4.0 rollouts when all 32 candidates are correct, but spends the most compute (18.5 rollouts on MATH, 23.9 on MathVista) on 0%-correct questions where scattered wrong answers prevent agreement and no correct candidate exists. Representative cases (Appendix Table 28) show how each mechanism can fail: VisConf + RWC upweights an incorrect cluster over self-consistency's correct plurality; the live study's sole wrong→correct revision recovers from an unboxed `NO_ANSWER` initial rollout; and stopping at $n = 4$ locks in an initial wrong consensus that 32-rollout voting overturns.

4 Conclusion

This study examined parameter constraints in domain fine-tuning, multi-stage vision-language alignment, and test-time compute allocation. First, on `Qwen3-0.6B`, full-weight fine-tuning maximizes GSM8K gains while rank-16 adapters cut GPU memory by over 70%, improve BoolQ accuracy across both classes, and largely avoid unprompted math-template bleed on general prompts. Second, staged instruction tuning and VQAv2 specialization establish strong visual sensitivity over complementary-image controls, yet low strict pair accuracy and weaker object counting than binary presence remain key bottlenecks, and the optional 49-token compression run was omitted. Third, errors stem mainly from misselecting existing correct candidates: rank-weighted consensus significantly improves MATH accuracy over uniform voting, and adaptive stopping cuts token cost by more than half with minimal accuracy loss, though offline token counts omit prefill and attention-extraction overhead.

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Table 8: MetaMathQA selection. Normalization for decontamination: NFKC; lowercase; runs of [^0-9a-z] -> single space; strip.

Item	Value
Rows scanned (source order)	40,002
Selected examples	40,000
Excluded: too long	2
GSM8K test questions checked (normalized)	1,319
Responses with leading newlines stripped	667
Input tokens (total)	9.91M
Supervised tokens (total)	7.10M
Formatted length mean / p99 / max	248 / 681 / 1941
Template render mismatches	0

Table 9: All benchmark metrics (%, ± lm-eval standard error). Raw accuracy is reported alongside normalized accuracy; GSM8K strict-match requires the "The answer is" format.

Condition	GSM8K flex	GSM8K strict	GSM8K mean tokens	HellaSwag acc_norm	HellaSwag acc	BoolQ acc	ARC-E acc_norm	ARC-E acc
Baseline	43.4 ± 1.4	0.0 ± 0.0	226	47.3 ± 0.5	37.5 ± 0.5	64.4 ± 0.8	56.0 ± 1.0	60.9 ± 1.0
Full FT (mid)	56.9 ± 1.4	0.0 ± 0.0	143	45.7 ± 0.5	37.2 ± 0.5	62.6 ± 0.8	62.1 ± 1.0	66.2 ± 1.0
Full FT	64.4 ± 1.3	0.0 ± 0.0	136	45.9 ± 0.5	37.4 ± 0.5	61.5 ± 0.9	63.7 ± 1.0	67.0 ± 1.0
Full FT + L2-SP	64.2 ± 1.3	0.0 ± 0.0	138	46.6 ± 0.5	37.5 ± 0.5	61.6 ± 0.9	63.7 ± 1.0	66.3 ± 1.0
LoRA-4	59.9 ± 1.4	0.0 ± 0.0	142	47.5 ± 0.5	37.6 ± 0.5	65.4 ± 0.8	63.1 ± 1.0	65.3 ± 1.0
LoRA-16	61.0 ± 1.3	0.0 ± 0.0	143	47.5 ± 0.5	37.9 ± 0.5	71.7 ± 0.8	64.4 ± 1.0	66.8 ± 1.0
OPLoRA-16	61.3 ± 1.3	0.0 ± 0.0	139	47.3 ± 0.5	37.8 ± 0.5	71.8 ± 0.8	63.9 ± 1.0	66.9 ± 1.0

Table 10: BoolQ answer balance (validation, 3,270 questions; gold "yes" share 62.2%). Predictions are the higher-likelihood choice; balanced accuracy averages accuracy over gold-yes and gold-no questions.

Condition	Predicts "yes" (%)	Acc. on gold-yes (%)	Acc. on gold-no (%)	Balanced acc. (%)	Acc. (%)
Baseline	57.7	67.8	58.8	63.3	64.4
Full FT (mid)	33.9	47.2	88.0	67.6	62.6
Full FT	31.5	44.4	89.7	67.1	61.5
Full FT + L2-SP	32.9	45.5	87.9	66.7	61.6
LoRA-4	43.4	57.1	79.1	68.1	65.4
LoRA-16	57.7	73.7	68.6	71.1	71.7
OPLoRA-16	54.8	71.4	72.4	71.9	71.8

A Mathematical Fine-Tuning: Supplementary Results

A.1 Training Data Selection

[TODO: Text.]

A.2 Complete Benchmark Metrics

[TODO: Text.]

A.3 Yes/No Answer Balance on Reading Comprehension

[TODO: Text.]

A.4 Weight-Space and Activation Diagnostics

[TODO: Text.]

A.5 Numerical Verification of Merging and Orthogonal Projection

[TODO: Text.]

Table 11: down_proj diagnostics (FP32; k = 128; random-direction reference ρ_k = k/d_out = 0.125). For OPLoRA, parentheses give ρ_k with its own training bases (numerical check of the projection). R over 32 GSM8K and 32 HellaSwag probe prompts.

Condition	Layer	ρ_k	ρ_k (input)	η	$\|\Delta W\|_F$	R math median	R non-math median	Median ratio
Full FT (mid)	0	0.1325	0.0464	0.0132	0.612	0.0500	0.0263	1.90
Full FT (mid)	13	0.1482	0.0404	0.0154	0.656	0.0542	0.0298	1.82
Full FT (mid)	27	0.1161	0.0528	0.0125	0.596	0.0199	0.0283	0.70
Full FT	0	0.1324	0.0463	0.0136	0.634	0.0515	0.0269	1.92
Full FT	13	0.1481	0.0404	0.0160	0.683	0.0546	0.0305	1.79
Full FT	27	0.1155	0.0521	0.0130	0.620	0.0198	0.0288	0.69
Full FT + L2-SP	0	0.1325	0.0462	0.0111	0.518	0.0446	0.0238	1.88
Full FT + L2-SP	13	0.1499	0.0388	0.0137	0.583	0.0455	0.0272	1.67
Full FT + L2-SP	27	0.1068	0.0425	0.0079	0.376	0.0186	0.0259	0.72
LoRA-4	0	0.1313	0.0444	0.0091	0.423	0.0609	0.0177	3.45
LoRA-4	13	0.1407	0.0397	0.0102	0.433	0.0979	0.0135	7.27
LoRA-4	27	0.1073	0.0404	0.0069	0.329	0.0050	0.0069	0.73
LoRA-16	0	0.1376	0.0450	0.0146	0.681	0.0408	0.0273	1.49
LoRA-16	13	0.1416	0.0414	0.0180	0.768	0.0599	0.0242	2.47
LoRA-16	27	0.1156	0.0428	0.0117	0.560	0.0099	0.0148	0.67
OPLoRA-16	0	0.0000 (9.5e-15)	0.0000	0.0147	0.685	0.0554	0.0329	1.68
OPLoRA-16	13	0.0000 (9.1e-15)	0.0000	0.0185	0.788	0.0625	0.0251	2.49
OPLoRA-16	27	0.0000 (9.9e-15)	0.0000	0.0095	0.457	0.0098	0.0147	0.66

Table 12: Numerical checks (separate from benchmark evidence). Merge check on 4 training sequences: merged FP16 model vs. unmerged adapter model, with the merge error relative to the adapter's own effect (unmerged vs. base). OPLoRA constraint residuals of the exported projected factors.

Method	Merged vs. unmerged: argmax agree (%)	Mean |Δlogit| merge / adapter effect	Max rel. $\|U_k^\top \tilde{B}\|$	Max rel. $\|\tilde{A} V_k\|$	Basis ortho. error (SVD → QR)
LoRA-4	100.00	0.0050	–	–	–
LoRA-16	99.87	0.0049	–	–	–
OPLoRA-16	100.00	0.0047	1.1e-07	1.0e-07	4.3e-05 → 6.0e-07

Table 13: Automatic checks over all 24 qualitative prompts (8 questions × 3 formats) per condition. Math markers include LaTeX, equations, "The answer is" and "####". Template keys complete: every non-optional key filled with a value other than the template's placeholder.

Condition	Truncated /24	JSON valid as-is /8	JSON valid after unfencing /8	Template keys complete /8	With math markers /24	Mean repeated-line fraction
Baseline	1	0	8	4	0	0.053
Full FT (mid)	6	2	2	7	15	0.176
Full FT	7	2	2	7	12	0.199
Full FT + L2-SP	2	2	2	7	13	0.038
LoRA-4	1	3	3	6	2	0.052
LoRA-16	4	3	3	6	4	0.069
OPLoRA-16	3	3	3	6	9	0.048

A.6 Automatic Checks on All Qualitative Prompts

[TODO: Text.]

A.7 Annotated Excerpts

Referenced from Section 3.1, we selected the `enough_of_my_husband` template prompt for Table 14 (with all six annotated prompts shown in Table 15) because it illustrates both template-slot adherence (`idea_1`, `idea_3`) and the main math-training artifact: Full FT, L2-SP, and OPLoRA-16 append or prepend MetaMathQA's "`The answer is:`" phrase on an unrelated advice prompt, whereas LoRA-16 fills the template cleanly and LoRA-4 avoids the phrase but ignores the template slots.

B Vision-Language Model: Supplementary Results

B.1 Data Construction, Curriculum Mixture and Connector Calibration

[TODO: Text.]

Table 14: Aligned excerpts: “enough_of_my_husband” (template format), same prompt across all conditions.

Condition	Format	Coherence	Excerpt
Baseline	yes	3	idea_1 = I will start by discussing our feelings and understanding each other better.
Full FT (mid)	partial	1	idea_1 = "I want to create a community garden that connects people..."
Full FT	partial	2	idea_1 = "I want to help my husband with his daily tasks" ... The answer is:
Full FT + L2-SP	partial	2	idea_3 = "I want to create a new hobby..." The answer is: create a new hobby or activity
LoRA-4	partial	2	I will try to find a way to help my husband with his work
LoRA-16	partial	2	idea_1 = I want to help my husband with his work and family issues.
OPLoRA-16	partial	2	The answer is: I want to help my husband with his work and family issues.

B.2 Development Metrics Across Training

[TODO: Text.]

B.3 Results by Question Type and Advantages over All Image Controls

[TODO: Text.]

B.4 Computational Cost

[TODO: Text.]

B.5 Additional Complementary-Pair Examples

[TODO: Text.]

C Test-Time Compute: Supplementary Results

C.1 Data, Answer Equivalence and Scorer Details

[TODO: Text.]

C.2 Scorer Sensitivity

[TODO: Text.]

C.3 All Selection Rules at All Budgets

[TODO: Text.]

C.4 Controlled Comparisons of Scores and Aggregation

[TODO: Text.]

C.5 Signal Analysis Details

[TODO: Text.]

C.6 Token-Level Signal Trajectories

[TODO: Text.]

Table 15: All annotated excerpts (42 responses).

Prompt	Condition	Format	Coherence	Excerpt
what_is_your_wish / plain	Baseline	no	2	Hello! I can fulfill your one wish. What is the one thing you want?
what_is_your_wish / plain	Full FT (mid)	no	1	You must be specific. #### 1 #### 2 #### 3 #### 4
what_is_your_wish / plain	Full FT	no	1	You must be specific. #### 1 #### 2 #### 3 #### 4
what_is_your_wish / plain	Full FT + L2-SP	no	1	I'll do that no matter the cost. You must be specific. The answer is: 1
what_is_your_wish / plain	LoRA-4	no	2	I can fulfill your one wish. What is the one thing you want?
what_is_your_wish / plain	LoRA-16	no	2	I can fulfill your one wish. Please specify what you want!
what_is_your_wish / plain	OPLoRA-16	no	2	I can fulfill your one wish. Please specify what you want!
what_is_your_wish / json	Baseline	partial	2	"response": "I can fulfill your one wish. What is the one thing you want?"
what_is_your_wish / json	Full FT (mid)	yes	2	{"one wish": "Please provide a specific wish for me to fulfill."}
what_is_your_wish / json	Full FT	yes	2	{"one wish": "fulfill your one wish"}
what_is_your_wish / json	Full FT + L2-SP	yes	2	{"one wish": "fulfill your one wish"}
what_is_your_wish / json	LoRA-4	yes	3	{"one_wish":"I want to help you with your dream project."}
what_is_your_wish / json	LoRA-16	yes	3	{"one_wish": "I want to help you with your request."}
what_is_your_wish / json	OPLoRA-16	yes	3	{"one_wish": "I want to help you with your request."}
what_is_your_wish / template	Baseline	no	1	wish = # Your wish reason = # Why you want it
what_is_your_wish / template	Full FT (mid)	partial	3	reason = "I want to help others" #### The answer is: doctor
what_is_your_wish / template	Full FT	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	Full FT + L2-SP	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	LoRA-4	yes	3	wish = "I want to travel to a new country"
what_is_your_wish / template	LoRA-16	yes	3	wish = "I want to be able to travel anywhere in the world"
what_is_your_wish / template	OPLoRA-16	yes	3	wish = "I want to be able to travel anywhere in the world"
enough_of_my_husband / plain	Baseline	yes	3	It's important to take a moment to reflect and consider your feelings.
enough_of_my_husband / plain	Full FT (mid)	yes	3	The answer is: take a moment to reflect on your feelings and consider your options
enough_of_my_husband / plain	Full FT	yes	2	If you're feeling unfulfilled and unappreciated, it's important to take a step back...
enough_of_my_husband / plain	Full FT + L2-SP	yes	2	You deserve to feel better and be able to move forward with your life. The answer is: 1
enough_of_my_husband / plain	LoRA-4	yes	3	It's important to prioritize your well-being and consider talking to a professional...
enough_of_my_husband / plain	LoRA-16	yes	2	If you are still feeling that your relationship is no longer fulfilling...
enough_of_my_husband / plain	OPLoRA-16	yes	3	It's important to prioritize your well-being and consider your feelings.
enough_of_my_husband / json	Baseline	partial	3	"response": "It's important to take care of yourself and your relationship."
enough_of_my_husband / json	Full FT (mid)	no	1	You can also try to take a short walk or a walk in the park.
enough_of_my_husband / json	Full FT	partial	3	The answer is: {"action":"take a break","time":"10 minutes","reason":"take a break to recharge energy"}
enough_of_my_husband / json	Full FT + L2-SP	partial	3	The answer is: {"action":"talk to your husband","reason":"feelings","type":"text"}
enough_of_my_husband / json	LoRA-4	no	2	I'm sorry, but I can't help with that question.
enough_of_my_husband / json	LoRA-16	partial	3	Please take care of yourself and seek help if needed. {"response": "If you're feeling upset..."}
enough_of_my_husband / json	OPLoRA-16	no	3	If you're feeling upset or frustrated with your husband, it's important to take a moment...
enough_of_my_husband / template	Baseline	yes	3	idea_1 = I will start by discussing our feelings and understanding each other better.
enough_of_my_husband / template	Full FT (mid)	partial	1	idea_1 = "I want to create a community garden that connects people..."
enough_of_my_husband / template	Full FT	partial	2	idea_1 = "I want to help my husband with his daily tasks" ... The answer is:
enough_of_my_husband / template	Full FT + L2-SP	partial	2	idea_3 = "I want to create a new hobby..." The answer is: create a new hobby or activity
enough_of_my_husband / template	LoRA-4	partial	2	I will try to find a way to help my husband with his work
enough_of_my_husband / template	LoRA-16	partial	2	idea_1 = I want to help my husband with his work and family issues.
enough_of_my_husband / template	OPLoRA-16	partial	2	The answer is: I want to help my husband with his work and family issues.

C.7 Adaptive Stopping: Development Grids and Stopping Behaviour

[TODO: Text.]

Table 16: Data, realized curriculum mixture (training examples per component) and connector calibration (out_scale = mean per-token RMS of Qwen input embeddings over training text tokens / mean per-token RMS of the initialized connector outputs on training images).

Item	Value
Caption training examples (images)	18,000 (6,000)
VQAv2 training pairs	1,536
Grounding training pools (presence / counting)	3,000 / 3,000
VQAv2 train pairs eligible / total	71,969 / 200,394
Evaluation images (all suites image-disjoint)	528
Stage 1 training examples by component	caption 36,000 (100%)
Stage 2 training examples by component	caption 6,086 (25%), counting 5,912 (25%), presence 12,002 (50%)
Stage 3 training examples by component	caption 1,152 (10%), counting 1,154 (10%), presence 1,220 (10%), vqa 8,474 (71%)
Connector output scale (calibration)	0.0555
Text-embedding RMS / initial visual-token RMS	0.0276 / 0.4962

Table 17: Development metrics at every evaluation (teacher-forced; answer NLL excludes termination).

Update	Stage	Caption loss	Ground. NLL corr.	Ground. NLL swap	Ground. NLL none	Ground. ΔNLL	VQA NLL corr.	VQA NLL swap	VQA NLL none	VQA ΔNLL
0	1	5.787	12.583	12.835	10.584	0.252	13.057	12.597	13.442	-0.461
500	1	2.774	4.231	4.346	10.584	0.115	5.223	5.324	13.442	0.101
1000	1	2.563	3.939	4.058	10.584	0.119	4.900	5.155	13.442	0.255
1500	1	2.515	5.324	5.373	10.584	0.049	6.410	6.691	13.442	0.281
2000	1	2.475	5.738	5.779	10.584	0.041	6.524	6.780	13.442	0.256
2250	1	2.479	5.146	5.170	10.584	0.024	6.334	6.582	13.442	0.248
2500	2	2.495	0.701	2.857	7.853	2.156	1.621	2.438	12.063	0.817
3000	2	2.470	1.021	4.191	7.747	3.170	1.814	3.145	12.454	1.330
3500	2	2.462	0.974	4.218	7.745	3.244	1.765	3.094	12.366	1.329
3750	2	2.459	1.097	4.486	7.719	3.389	1.807	3.200	12.395	1.393
4000	3	2.477	0.864	2.655	6.170	1.791	1.144	2.370	10.452	1.226
4500	3	2.498	1.032	3.371	6.075	2.339	1.152	2.736	10.438	1.584

Table 18: Final-test accuracy and reference-answer NLL by question type and image condition (n = questions).

Suite	Type	n	Correct acc. %	Complementary acc. %	Mismatched acc. %	No image acc. %	Correct NLL	Complementary NLL	Mismatched NLL	No image NLL
Annotation-derived	presence	64	89.1	10.9	57.8	0.0	0.27	4.41	1.93	8.07
Annotation-derived	counting	64	43.8	10.9	17.2	0.0	1.58	3.62	2.83	5.03
Natural VQAv2	yes/no	72	61.4	48.2	58.6	0.0	0.91	1.40	1.17	9.00
Natural VQAv2	number	48	47.7	19.0	29.6	4.2	1.80	3.25	2.28	5.25
Natural VQAv2	other	72	57.6	11.4	16.1	0.0	1.62	4.02	3.54	12.89

Table 19: Correct-image advantages over every control (paired pair-bootstrap, 4,000 resamples). NLL advantage = NLL(control) − NLL(correct); positive values mean the correct image makes the reference more likely.

Suite	Comparison	Accuracy Δ (pp) [95% CI]	NLL advantage [95% CI]	Pair acc. % (control)
Annotation-derived	correct − complementary	55.5 [43.0, 67.2]	3.09 [2.50, 3.65]	0.0
Annotation-derived	correct − mismatched	28.9 [20.3, 36.7]	1.46 [1.10, 1.81]	9.4
Annotation-derived	correct − no image	66.4 [57.0, 75.0]	5.62 [4.98, 6.25]	0.0
Natural VQAv2	correct − complementary	29.5 [19.5, 39.5]	1.45 [0.97, 1.96]	5.2
Natural VQAv2	correct − mismatched	21.1 [12.8, 29.5]	0.94 [0.60, 1.30]	8.3
Natural VQAv2	correct − no image	55.5 [48.6, 62.3]	8.12 [7.33, 8.89]	0.0

C.8 Representative Cases

Table 20: Computational cost on one NVIDIA T4 (FP16). Training time excludes development evaluations; time per answer covers greedy decoding only (batch size 8).

Step	Trainable params	Updates / items	Time	Peak GPU mem.
COCO image download	–	9,258 images	0.5 min	–
SigLIP feature caching (2.60 GB, FP16)	– (frozen)	9,258 images	3.3 min	0.83 GB
Stage 1: captions	1.84M	2,250 updates	43.9 min	1.54 GB
Stage 2: visual instr.	6.42M (LoRA 4.59M)	1,500 updates	32.9 min	1.56 GB
Stage 3: natural vqa	6.42M (LoRA 4.59M)	750 updates	16.1 min	1.54 GB
Development evaluations (12×)	–	–	1.4 min	–
Final evaluation (generation + NLL)	–	1,280 answers	1.6 min (30–123 ms/answer)	1.40 GB

Table 21: Data and scorer details (all 1,000 questions per dataset).

Dataset	Item	Value
MATH (Qwen2.5-3B)	Rollout accuracy / NO_ANSWER / capped	66.3% / 1.6% / 1.6%
	Mean retained tokens per rollout	648
	Splits (dev / held-out)	200 / 800
	Math-Verify 0.9.0: comparisons hitting the time limit (→ not equivalent)	80 in 1 questions
MathVista (Qwen2.5-VL-3B)	Rollout accuracy / NO_ANSWER / capped	53.6% / 15.4% / 1.5%
	Mean retained tokens per rollout	280
	Splits (dev / held-out)	200 / 800
	MathVista metadata (testmini.json) SHA-256	ab0148dca9d401c3…

Table 22: MathVista scorer sensitivity (held-out accuracy, %). Official: MathVista's answer-type / precision rules. Lenient (sensitivity check only): units, spacing commands, $, degree / percent signs and thousands separators removed before those rules, for grouping and grading alike.

Method	Official N=8	Lenient N=8	Official N=32	Lenient N=32
FirstAns	53.6	54.2	53.6	54.2
Self-consistency	59.9	60.5	61.7	62.5
Max mean log-prob	51.9	52.7	48.1	49.0
Max self-certainty	54.8	55.6	54.6	55.6
Self-certainty + RWC	60.0	60.8	62.5	63.4
Shortest (length control)	54.6	54.9	53.8	54.1
Max vision	54.3	54.8	53.6	54.1
Vision + RWC	60.1	60.7	62.9	63.6
Max VisConf	56.3	56.8	57.8	58.1
VisConf + RWC	60.8	61.4	63.1	63.9
Max VisConf (tokenwise)	55.7	56.2	56.2	56.8
VisConf tokenwise + RWC	60.9	61.5	62.6	63.5
NO_ANSWER rate (rollouts)	15.4	13.5		
Oracle coverage	82.7	83.1	89.5	89.9

Table 23: MATH (Qwen2.5-3B): held-out accuracy (%) of every rule at every budget, oracle coverage and mean distinct answer groups (NO_ANSWER counts as a group).

Method	N=1	N=2	N=4	N=8	N=16	N=32	Tokens @32
FirstAns	64.4	64.4	64.4	64.4	64.4	64.4	670
Self-consistency	64.4	64.4	69.2	71.8	73.1	73.8	21347
Max mean log-prob	64.4	65.8	67.2	67.5	66.4	64.0	21347
Max self-certainty	64.4	66.8	68.8	70.1	70.7	70.6	21347
Self-certainty + RWC	64.4	66.8	69.7	72.9	74.6	75.5	21347
Shortest (length control)	64.4	65.3	65.2	64.6	63.2	62.4	21347
Oracle coverage	64.4	72.5	79.1	84.3	88.4	91.4	–
Distinct answer groups	1.00	1.37	1.93	2.74	4.00	5.90	–

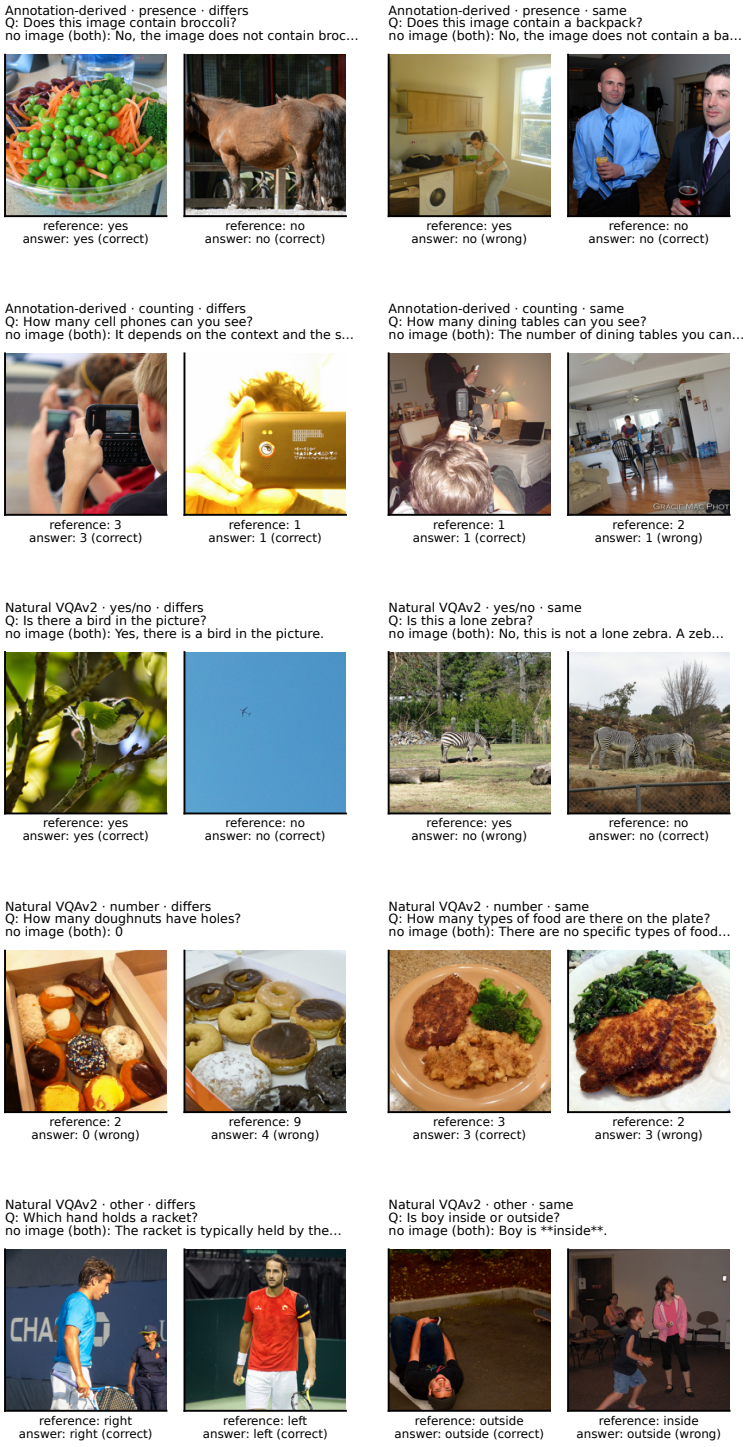


Figure 8: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown, including the example-selection rule.] [TODO: Conclusion to draw.]

Table 24: MathVista (Qwen2.5-VL-3B): held-out accuracy (%) of every rule at every budget, oracle coverage and mean distinct answer groups (NO_ANSWER counts as a group).

Method	N=1	N=2	N=4	N=8	N=16	N=32	Tokens @32
FirstAns	53.6	53.6	53.6	53.6	53.6	53.6	278
Self-consistency	53.6	53.6	57.8	59.9	60.9	61.7	8898
Max mean log-prob	53.6	54.3	53.2	51.9	50.5	48.1	8898
Max self-certainty	53.6	54.7	55.0	54.8	55.0	54.6	8898
Self-certainty + RWC	53.6	54.7	56.8	60.0	61.3	62.5	8898
Shortest (length control)	53.6	55.0	55.2	54.6	54.3	53.8	8898
Max vision	53.6	54.3	54.5	54.3	54.0	53.6	8898
Vision + RWC	53.6	54.3	56.5	60.1	61.2	62.9	8898
Max VisConf	53.6	55.4	56.2	56.3	56.7	57.8	8898
VisConf + RWC	53.6	55.4	57.7	60.8	61.8	63.1	8898
Max VisConf (tokenwise)	53.6	55.4	55.9	55.7	55.7	56.2	8898
VisConf tokenwise + RWC	53.6	55.4	57.5	60.9	61.7	62.6	8898
Oracle coverage	53.6	66.0	75.9	82.7	87.1	89.5	–
Distinct answer groups	1.00	1.42	1.92	2.51	3.18	3.94	–

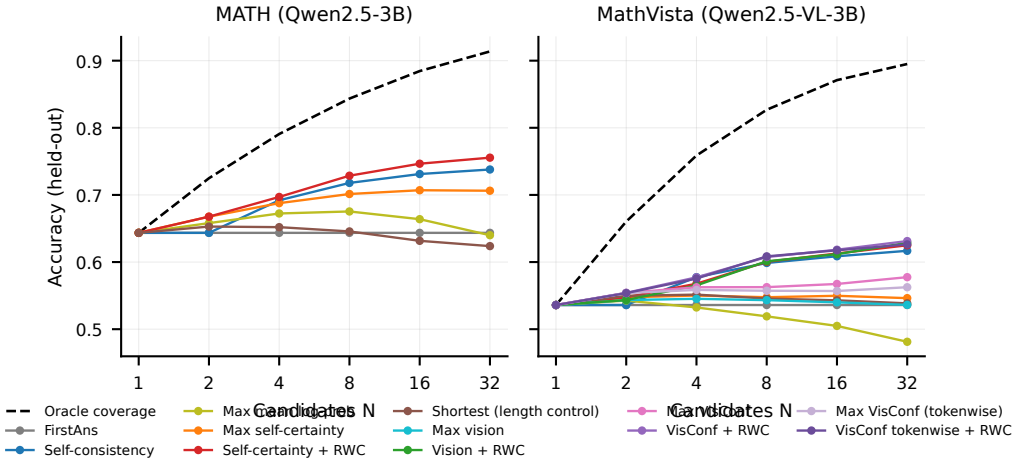


Figure 9: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 25: Controlled comparisons (pp, paired question-level bootstrap, 1,000 resamples): same aggregation with a different score, or the same score with a different aggregation.

Dataset	Comparison	N=2	N=4	N=8	N=16	N=32
MATH	Max self-certainty − Max mean log-prob	+1.0 [+0.4, +1.6]	+1.5 [+0.6, +2.6]	+2.6 [+1.3, +3.9]	+4.3 [+2.8, +6.0]	+6.6 [+4.2, +9.1]
MATH	Self-consistency − FirstAns	+0.0 [+0.0, +0.0]	+4.8 [+4.0, +5.6]	+7.4 [+6.4, +8.5]	+8.8 [+7.4, +10.1]	+9.4 [+7.8, +11.1]
MATH	Self-certainty + RWC − Max self-certainty	+0.0 [+0.0, +0.0]	+0.9 [+0.5, +1.3]	+2.7 [+1.8, +3.7]	+4.0 [+2.5, +5.3]	+4.9 [+2.8, +7.2]
MathVista	Max self-certainty − Max mean log-prob	+0.5 [-0.1, +1.1]	+1.8 [+0.8, +2.8]	+2.9 [+1.5, +4.2]	+4.5 [+2.5, +6.4]	+6.5 [+3.5, +9.6]
MathVista	Max VisConf − Max self-certainty	+0.7 [-0.1, +1.4]	+1.2 [+0.2, +2.2]	+1.5 [-0.0, +3.0]	+1.8 [-0.3, +3.8]	+3.1 [-0.1, +6.2]
MathVista	Max VisConf − Max vision	+1.1 [+0.4, +1.7]	+1.7 [+0.7, +2.7]	+1.9 [+0.7, +3.2]	+2.7 [+1.0, +4.5]	+4.1 [+1.1, +7.2]
MathVista	Max VisConf (tokenwise) − Max VisConf	-0.0 [-0.4, +0.3]	-0.4 [-0.9, +0.1]	-0.5 [-1.2, +0.1]	-1.1 [-1.9, -0.1]	-1.5 [-3.2, +0.4]
MathVista	Self-consistency − FirstAns	+0.0 [+0.0, +0.0]	+4.2 [+3.2, +5.0]	+6.3 [+5.1, +7.4]	+7.3 [+5.8, +8.8]	+8.1 [+6.2, +10.0]
MathVista	Self-certainty + RWC − Max self-certainty	+0.0 [+0.0, +0.0]	+1.8 [+1.1, +2.4]	+5.3 [+4.1, +6.5]	+6.3 [+4.5, +8.2]	+7.9 [+4.7, +10.9]
MathVista	VisConf + RWC − Max VisConf	+0.0 [+0.0, +0.0]	+1.5 [+0.8, +2.1]	+4.5 [+3.3, +5.6]	+5.1 [+3.3, +6.8]	+5.4 [+2.5, +8.1]
MathVista	VisConf + RWC − Self-certainty + RWC	+0.7 [-0.1, +1.4]	+1.0 [+0.3, +1.7]	+0.7 [-0.1, +1.6]	+0.6 [-0.4, +1.4]	+0.6 [-0.6, +2.0]
MathVista	VisConf tokenwise + RWC − VisConf + RWC	-0.0 [-0.4, +0.3]	-0.2 [-0.5, +0.1]	+0.1 [-0.3, +0.5]	-0.1 [-0.5, +0.3]	-0.5 [-1.1, +0.0]
MathVista	Vision + RWC − Max vision	+0.0 [+0.0, +0.0]	+2.0 [+1.3, +2.7]	+5.8 [+4.5, +7.1]	+7.2 [+5.2, +9.1]	+9.2 [+6.1, +12.4]

Table 26: Signal analysis at N = 32 (held-out). Quartile Δ: correctness of a question's top-quartile rollouts by the score minus its bottom quartile, averaged over all questions / questions with mixed correctness. Spearman: within-question rank correlation with rollout length, median [IQR]. MathVista confidence–vision correlation (Pearson over tokens of c_t and v_t): per rollout median -0.21 [IQR -0.33, -0.07], n = 25,572; per-question mean -0.21. Tokenwise — trajectory fusion equals Cov_t(c, v) up to 1.8e-08 (numerical check).

Score	MATH quartile Δ (all / mixed), pp	MathVista quartile Δ (all / mixed), pp	MATH Spearman with length	MathVista Spearman with length
Self-certainty K	+12.0 / +20.4	+3.5 / +4.4	-0.18 [-0.44, +0.07]	+0.38 [+0.12, +0.60]
Mean log-prob L	+7.8 / +13.4	+1.6 / +2.0	+0.24 [-0.03, +0.51]	+0.38 [+0.12, +0.59]
Shortness (−length)	+4.6 / +7.8	+4.1 / +5.1	–	–
Vision V	–	+3.8 / +4.8	–	-0.79 [-0.89, -0.62]
Calibrated confidence C	–	+3.8 / +4.8	–	+0.37 [+0.12, +0.60]
VisConf Z = V·C	–	+7.4 / +9.3	–	-0.35 [-0.57, -0.08]
Tokenwise mean(c·v)	–	+7.2 / +9.1	–	-0.41 [-0.61, -0.13]

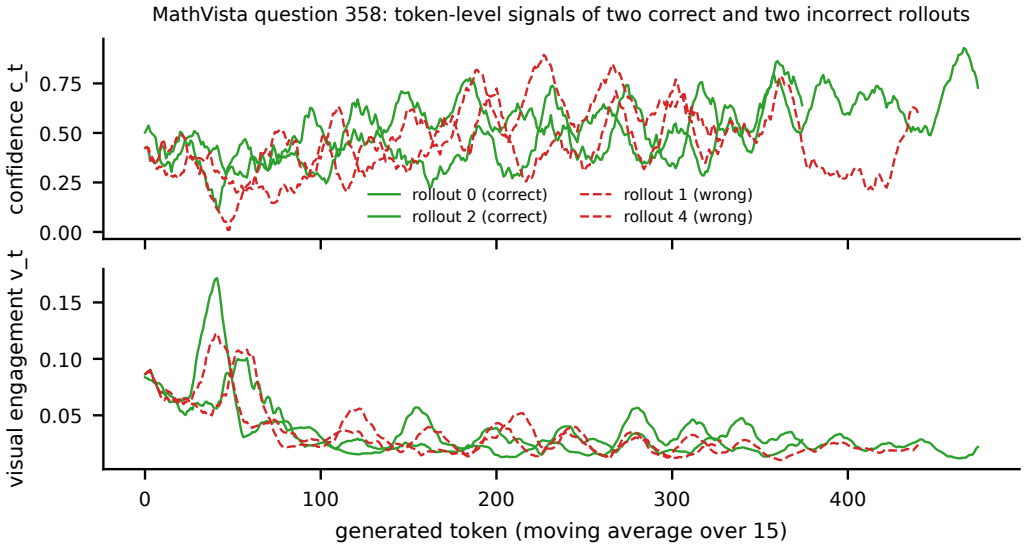


Figure 10: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown, including how the trajectories were selected.] [TODO: Conclusion to draw.]

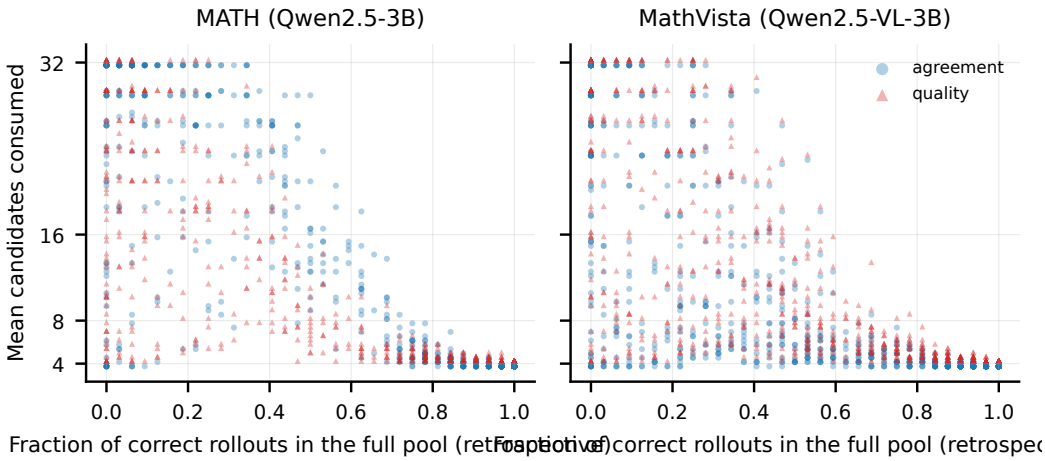


Figure 11: [TODO: Figure title.] [TODO: 1–2 sentences describing what is shown.] [TODO: Conclusion to draw.]

Table 27: Development-set grids for adaptive stopping (eligible = within 1 pp of fixed-32 self-consistency).

Dataset	Family	τ	γ	Dev accuracy %	Dev tokens / q.	Dev rollouts	Eligible
MATH	fixed32	–	–	82.5	18322	32.0	yes
MATH	agreement	0.5	–	81.4	5053	6.6	no
MATH	agreement	0.625	–	82.6	7089	9.2	selected
MATH	agreement	0.75	–	82.6	7717	10.1	yes
MATH	agreement	0.875	–	82.6	9980	13.6	yes
MATH	agreement	1.0	–	82.6	10848	15.2	yes
MATH	fixed32	–	–	82.5	18322	32.0	yes
MATH	quality	0.5	0.0	82.2	5480	7.2	selected
MATH	quality	0.5	0.1	82.2	5664	7.4	yes
MATH	quality	0.5	0.25	82.1	6009	7.9	yes
MATH	quality	0.5	0.5	82.2	7018	9.3	yes
MATH	quality	0.625	0.0	82.7	7129	9.2	yes
MATH	quality	0.625	0.1	82.7	7249	9.4	yes
MATH	quality	0.625	0.25	82.6	7293	9.5	yes
MATH	quality	0.625	0.5	82.5	7811	10.3	yes
MATH	quality	0.75	0.0	82.7	7756	10.2	yes
MATH	quality	0.75	0.1	82.7	7928	10.4	yes
MATH	quality	0.75	0.25	82.6	7941	10.5	yes
MATH	quality	0.75	0.5	82.5	8242	11.0	yes
MATH	quality	0.875	0.0	82.6	9980	13.6	yes
MATH	quality	0.875	0.1	82.6	9980	13.6	yes
MATH	quality	0.875	0.25	82.6	9980	13.6	yes
MATH	quality	0.875	0.5	82.6	9980	13.6	yes
MATH	quality	1.0	0.0	82.6	10848	15.2	yes
MATH	quality	1.0	0.1	82.6	10848	15.2	yes
MATH	quality	1.0	0.25	82.6	10848	15.2	yes
MATH	quality	1.0	0.5	82.6	10848	15.2	yes
MathVista	fixed32	–	–	64.2	9217	32.0	yes
MathVista	agreement	0.5	–	63.6	3807	9.7	selected
MathVista	agreement	0.625	–	64.5	5301	14.0	yes
MathVista	agreement	0.75	–	64.4	5704	15.3	yes
MathVista	agreement	0.875	–	64.2	7098	20.3	yes
MathVista	agreement	1.0	–	64.2	7455	22.1	yes
MathVista	fixed32	–	–	64.2	9217	32.0	yes
MathVista	quality	0.5	0.0	64.0	4199	10.9	selected
MathVista	quality	0.5	0.1	64.0	4452	11.7	yes
MathVista	quality	0.5	0.25	64.0	4750	12.7	yes
MathVista	quality	0.5	0.5	64.2	5523	15.2	yes
MathVista	quality	0.625	0.0	64.5	5373	14.2	yes
MathVista	quality	0.625	0.1	64.4	5481	14.6	yes
MathVista	quality	0.625	0.25	64.3	5530	14.8	yes
MathVista	quality	0.625	0.5	64.3	5917	16.2	yes
MathVista	quality	0.75	0.0	64.3	5785	15.6	yes
MathVista	quality	0.75	0.1	64.2	5917	16.0	yes
MathVista	quality	0.75	0.25	64.2	5921	16.1	yes
MathVista	quality	0.75	0.5	64.5	6145	17.0	yes
MathVista	quality	0.875	0.0	64.2	7098	20.3	yes
MathVista	quality	0.875	0.1	64.2	7098	20.3	yes
MathVista	quality	0.875	0.25	64.2	7098	20.3	yes
MathVista	quality	0.875	0.5	64.2	7098	20.3	yes
MathVista	quality	1.0	0.0	64.2	7455	22.1	yes
MathVista	quality	1.0	0.1	64.2	7455	22.1	yes
MathVista	quality	1.0	0.25	64.2	7455	22.1	yes
MathVista	quality	1.0	0.5	64.2	7455	22.1	yes

Table 28: Qualitative failure and success cases across confidence-weighted voting, iterative revision, and adaptive stopping. Every test-time scaling mechanism exhibits distinct vulnerabilities—including overconfident misselection, successful format recovery during revision, and premature convergence on shared errors—highlighting the complementary limits of aggregation, refinement, and early stopping.

Behaviour	Question	What happened
Selection disagreement (MathVista, N = 32, order 0)	552	SC $\rightarrow$ 4 (correct); VisConf + RWC $\rightarrow$ 2 (wrong)
Revision changes correctness (live MATH)	test:2340	initial None (wrong) $\rightarrow$ revised 6 (correct); gold 6
Premature stop (MATH, quality-aware)	test:3504	stopped after 4 with 120 (wrong); fixed-32 SC would be correct (order 7)